Name: Student ID:

Today's data is on house pricing in Los Angeles! We have access to data on all of the houses that sold in four cities around west LA in the course of a single month. The `LA` data frame is available in the Files section of bCourses.

1. Our eventual goal is to predict the price of a home based on a few attributes of that home. But first, create a scatterplot examining the relationship between square footage and price.

- Write the code you used below.

- Sketch the plot below.

2. Create a scatter plot examining the relationship between square footage *squared* (raised to the second power) and price. Sketch the plot below.

3. Add a columns onto the `LA` dataset, `log_sqft`, which measures *log* square footage. You may use *log base 10*. Save the new dataset back into itself.

4. Create another scatterplot examining the relationship between *log* square footage and price.

- Sketch the plot below.

5. Explain in at least two sentences whether the logged predictor variable or the regular predictor appears more suitable to predict price with using a linear model.

6. Fit three linear models

- one which predicts price with square footage;
- one which predicts price with the square footage raised to the second power (square footage squared);
- one which predicts price with log square footage.

You do not need to copy the code below, but, using the `broom` library, find and report the $R^2$ for both models below.

| Model | $R^2$ |
|---|---|
| `price ~ sqft` | |
| `price ~ sqft^2` | |
| `price ~ log_sqft` | |

7. In one to two sentences, explain whether the results you have line up with your explanation in **Question 5**.

Recently the University of California purchased a new house to serve as the residence of the university President and to host university functions. The address of the house is 2821 Claremont Blvd in Berkeley. You can find a listing for the house on the Internet!

8. Use the linear model with the highest $R^2$ from above to predict the sale price in (USD) of this house.

9. Was your model an under- or over-estimate? Why do you think this is? Answer in at least two sentences.